# Logistics Operations Analysis

Capstone Project

Data Analysis using SQL & Excel

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# Project Overview

- **Business Context**
  The project analyzes the operations of a logistics and trucking company, focusing on revenue, fleet efficiency, fuel costs, delivery performance, and safety.
- **Dataset**
  The dataset contains operational data covering loads, trips, drivers, trucks, fuel purchases, delivery events, maintenance, customers, routes, and safety incidents.
- **Objective**
  To analyze the data and identify key operational patterns that can support better business decisions.

# Data Preparation & Tools

- **Data Preparation**
- Reviewed and validated the available datasets
- Checked for missing values, duplicates, and data consistency
- Retained valid records with missing IDs where appropriate
- Documented identified data-quality issues

- **Tools Used**
- Microsoft Excel — data review and validation
- SQL Server — data querying and analysis
- PowerPoint — presentation of findings

# Key Operational Metrics

- **85,410** loads processed
- **$262.53M** recorded revenue
- **$3,073.71** average revenue per load
- **122.16M miles** travelled
- **18.95M gallons** of fuel used
- **$95.59M** total fuel spending
- **6.50 MPG** average fuel efficiency
- **55.67%** overall on-time event rate

# Delivery Performance

- Total delivery events: **170,820**
- On-time events: **95,095**
- Overall on-time event rate: **55.67%**
- The results indicate an opportunity to improve service reliability

# Fleet & Fuel Efficiency

- Distance travelled: **122.16M miles**
- Fuel used: **18.95M gallons**
- Average fuel efficiency: **6.50 MPG**
- Fuel efficiency is an important area for cost control.

# Safety Performance

- Total incidents: **170**
- Preventable incidents: **64**
- At-fault incidents: **54**
- Injury incidents: **33**
- Total claims: **$2.65M**

# Revenue & Cost Analysis

- **Total recorded revenue:** $262.53M
- **Average revenue per load:** $3,073.71
- **Total fuel spending:** $95.59M
- **Average fuel purchase:** $486.62
- **Fuel spending represents a major operating cost**
- **Revenue and fuel costs should be monitored together when assessing operational performance**

# Key Insights

- The company generated substantial recorded revenue across **85,410 loads**.
- Fuel represents a significant operating cost.
- Fleet fuel efficiency averaged **6.50 MPG**.
- On-time performance was **55.67%** across delivery events.
- Safety incidents resulted in approximately **$2.65M** in claims.

# Recommendations & Conclusions

## Recommendations

- Monitor fuel consumption and MPG regularly.
- Investigate causes of delayed delivery events.
- Strengthen preventive safety measures.
- Track fuel and operational costs alongside revenue.

## Conclusion

- The analysis provides a clear view of revenue, fuel costs, fleet efficiency, delivery performance, and safety. These insights can help management identify opportunities to improve operational efficiency and reduce avoidable costs.